

Towards More General Video-based Deepfake Detection through Facial Component Guided Adaptation for Foundation Model

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Abstract

The current deep generative models have enabled the creation of synthetic facial images with remarkable photorealism, raising significant societal concerns over their potential misuse. Despite rapid advancements in the field of deepfake detection, developing an efficient and effective approach for the generalized deepfake detection of unseen forgery samples remains challenging. To address this challenge, we leverage the rich semantic priors of foundation models and propose a novel side-network-based decoder that extracts spatial and temporal cues using the CLIP image encoder for generalized video-based Deepfake detection. Additionally, we introduce Facial Component Guidance (FCG) to enhance spatial learning generalizability by encouraging the model to focus on key facial regions. By leveraging the generic features of a vision-language foundation model, our approach demonstrates promising generalizability on challenging Deepfake datasets while also exhibiting superiority in training data efficiency, parameter efficiency, and model robustness. The source code is available at: <https://github.com/aiiu-lab/DFD-FCG>

1. Introduction

With advancements in deep learning, generative models [15, 18, 20, 21, 34, 39, 41] have empowered the general public to create realistic synthetic images. In the wrong hands, these tools can easily be used to mislead viewers, incite social unrest and pose significant threats for personal reputation. In recent years, research on Deepfake detection has gained momentum to counter the growing risks posed by synthetic contents. However, many existing detection methods struggle to recognize Deepfakes, synthetic facial images and videos, generated by newer synthesis techniques [8, 12, 26, 27, 44, 56]. This underscores the importance of developing models with robust generalization capabilities that can accurately differentiate between real and Deepfake samples under unfamiliar scenarios.

Previous works [2, 4, 9, 12–14, 26, 44, 51–53] have

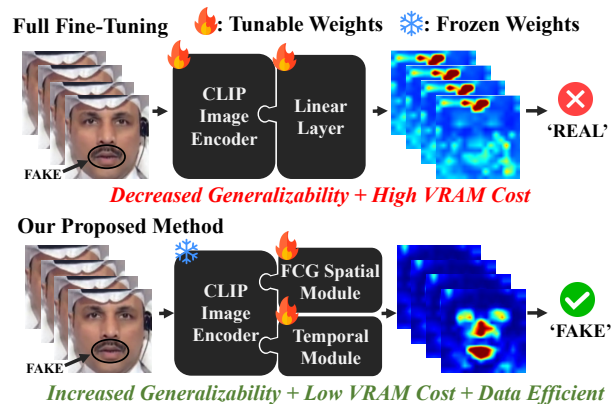


Figure 1. Our proposed method incorporates a Facial Component Guided (FCG) spatial module and a temporal module to adapt the foundational CLIP image encoder for the task of Deepfake detection. This approach enhances model generalizability by guiding it to focus on nuanced artifacts that appear in specific facial component regions rather than arbitrary areas.

addressed the generalization problem from various angles, such as utilizing specific facial features for Deepfake detection [12, 26], generating pseudo-Deepfake samples for self-supervised learning [2, 44], detecting temporal inconsistencies in Deepfake videos [55], and fine-tuning pre-trained expert models [13]. Despite these efforts, recent methods still face challenges, often learning dataset-specific cues that fail to detect unseen forgery samples.

Foundation Models (FMs) [23, 36, 38] – generic models trained on web-scale datasets – have demonstrated remarkable zero-shot performance and can be quickly adapted to various downstream tasks with few additional tuning cost. In terms of general synthetic image detection, Ojha et al. [35] has adapted the Contrastive Language-Image Pretraining (CLIP) [38] with an additional linear layer and achieved an impressive performance. However, its effectiveness in detecting Deepfakes (synthetic facial images/videos) remains largely unexplored. From our pilot study as shown in Figure 1, the full fine-tuned CLIP-based detector tends to focus its attention on the generic regions

and may potentially reduce its generalization performance. This raises an important question: How can we harness the rich inherent features of FMs to tackle the Deepfake detection task while benefiting from their advantages of fast adaptation and generalizability?

To address the above challenges, inspired by recent advancements in Parameter Efficient Fine-tuning Techniques (PEFT), we propose an efficient and generalizable framework based on a side-network-based decoder – comprising both spatial and temporal modules – to extract generic spatio-temporal representation based on the layer-wise features from the CLIP image encoder for the Deepfake detection task. Furthermore, we develop a Facial Component Guidance (FCG) mechanism to enhance the generalizability of the spatial module by focusing on key facial components during training. With extensive experiments and ablation studies, the proposed method outperforms other State-Of-The-Art (SOTA) methods in the cross-dataset evaluation. The results demonstrate the effectiveness of our approach.

Our main contributions are summarized as follows:

- We develop a novel side-network-based decoder with the spatial and temporal modules which not only extracts rich spatio-temporal features but also allows to fast adapt the CLIP image encoder to video-based Deepfake detection.
- The proposed FCG mechanism effectively enhances the spatial module’s generalization capability by directing the module’s attention to key facial regions.
- Our method outperforms other SOTA methods in cross-dataset evaluation. Extensive experiments further demonstrate the efficiency of our framework in terms of the number of training parameters and data, as well as its robustness against generic perturbations.

2. Related Work

In this section, we briefly review recent works on (1) Foundation Models (FMs) and Parameter Efficient Fine-tuning Techniques (PEFTs), and (2) Deepfake detection.

2.1. Foundation Model and Parameter Efficient Fine-tuning Techniques

Adapting large-scale FMs for downstream tasks has become the new gold standard in the field of modern machine learning. These models are highly adaptable for various downstream tasks with superior performance, characterized by their training on large-scale datasets through advanced methods, which yields rich and powerful feature extraction capabilities. Leveraging multi-modal information further enhances these models’ semantic understanding. Notable examples include CLIP [38] with its robust visual-text encodings, its expanded counterpart OpenCLIP [3], the image-generating Stable Diffusion [41], and the self-supervised DINOv2 [36]. These models accelerate downstream task development, as demonstrated by EVL [30] and

DIFT [45], which utilize FMs for video action recognition and image correspondence, respectively.

Adapting these models to specific tasks poses significant challenges, particularly due to their large model sizes, which has led to the development of partial fine-tuning strategies. The Low-Rank Adaptation (LoRA) [15] method adapts large language models by incorporating rank decomposition matrices into the Transformer structure, significantly reducing the number of trainable parameters and enabling efficient adaptation. The Visual-Prompt Tuning (VPT) [16] technique fine-tunes Vision Transformer (ViT) models using trainable “prompts” to minimize training overhead. The EXPRES [6] approach constructs downstream representations with residual learnable “output” tokens. The SAN [32] framework incorporates a streamlined side network into the static CLIP model for open-vocabulary semantic segmentation. Selecting the most effective adaptation strategy to maximize a model’s capabilities remains an important research focus.

2.2. Deepfake Detection

Advancements in deep generative methods have driven the development of diverse Deepfake detection techniques. Early approaches like Xception [42] performed well in intra-dataset evaluations but degraded significantly in cross-dataset scenarios. Later studies focused on artifacts that generative models fail to eliminate—for instance, FTCN [55] detects temporal inconsistencies, LipsForensics [12] targets unnatural lip movements, and EyesTel-IAll [11] identifies irregular pupil shapes.

The limited diversity of forgery samples in training datasets remains a key obstacle to robust model generalizability. To address this challenge, RealForensics [13] pre-trains a video backbone on large-scale lip-reading datasets before transferring it to the Deepfake detection task, while Yan et al. [53] simulate forgery features in the latent space to support learning a more generalizable decision boundary. A prominent line of research [2, 26, 29, 33, 44, 48, 54] synthesizes pseudo-Deepfake samples during training. However, we demonstrate in Sec. 9 that SBI [44] (the representative method of the category) performs well on GAN-generated samples but struggles with those produced by more recent diffusion-based models.

Further studies on video-based Deepfake detection aim to address challenges related to generalizability and memory efficiency. For example, AltFreezing [51] improves efficiency by alternately freezing the weights of spatial and temporal modules, and TALL-Swin [52] transforms video clips into predefined layouts to preserve spatial and temporal dependencies. Other notable efforts include leveraging the latent flow from StyleGAN [22] for enhanced generalizability, addressing the identity leakage problem [9], jointly performing binary Deepfake classification and manipula-

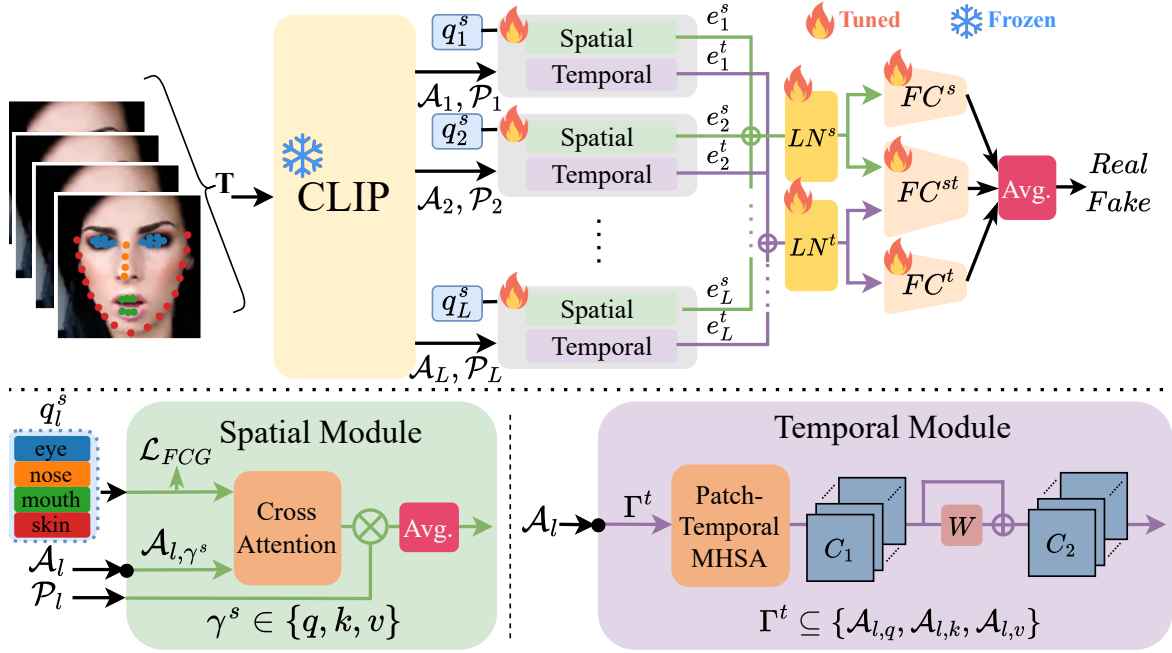


Figure 2. **Framework Overview:** Our method utilizes the CLIP image encoder to extract layer-wise features (attention attributes $\mathcal{A}_l$ and patch embeddings $\mathcal{P}_l$), which are then processed by the corresponding decoder block consist of temporal and spatial modules for parameter-efficient fine-tuning. The spatial module incorporates the FCG loss to focus on key facial parts in each frame to capture the Deepfake visual cues. The temporal module employs the Patch-Temporal Multi-Head Self-Attention to capture the temporal inconsistency of Deepfake videos. The $\bullet$ at the input of the two modules indicates where the required attributes are extracted based on the pre-configured settings. Ultimately, our framework aggregates the outputs of the spatial and temporal modules for the final prediction. The superscripts t , s , and st represent temporal related, spatial related, and spatio-temporal related components, respectively.

tion order detection [14], adjusting the composition of fake samples during training [29], and knowledge transfer from distorted face recovering [48]. However, the effectiveness of FMs in this field remains largely unexplored.

3. Method

Our approach focuses on effectively harnessing the inherent capabilities of foundation models to identify temporal inconsistencies and spatial manipulations in Deepfake videos. To this end, we develop a framework that leverages the rich representations of the pre-trained foundation model (i.e., CLIP image encoder) and integrates a side-network-based decoding block, comprising both specialized temporal and spatial modules, to capture these key characteristics for Deepfake detection, as illustrated in Figure 2. In the following sections, we first outline the model’s overall architecture (Section 3.1), and then delve into the details of the proposed spatial (Section 3.2) and temporal (Section 3.3) modules in the layer-wise decoders.

3.1. Overall Framework

Drawing inspiration from the works of [30, 31], we utilize the ViT-based CLIP image encoder to extract visual features

that our side-network-based spatial and temporal modules can leverage for Deepfake detection. To facilitate understanding of the proposed method, we first introduce several symbols and notations representing the system configuration: L denotes the number of transformer layers, P the number of patches, T the number of input frames, H the number of attention heads in multi-head self-attention, and D the feature dimension per head. Besides, **we follow the convention throughout the paper that the superscripts t , s , and st denote temporal, spatial, and spatio-temporal components, respectively.** (e.g., x^t , x^s , and x^{st} .)

For each transformer layer of the image encoder, we obtain patch embeddings $\mathcal{P}_l \in \mathbb{R}^{T \times P \times (H \times D)}$ and attention attributes $\mathcal{A}_{l,\gamma} \in \mathbb{R}^{T \times P \times H \times D}$, where $1 \leq l \leq L$ indicates the layer index, and $\gamma \in \{q, k, v\}$ specifies the attention attribute type (i.e., query, key and value). Please refer to Section 10 for more details. These layer-wise attributes are then processed by a layer-wise decoder block through the spatial and temporal modules, resulting in temporal and spatial embeddings e_l^t and e_l^s . We aggregate these layer-wise temporal and spatial embeddings, denoted as E^t and E^s , which are subsequently normalized using respective LayerNorm layers LN^t and LN^s . The framework employs three indepen-

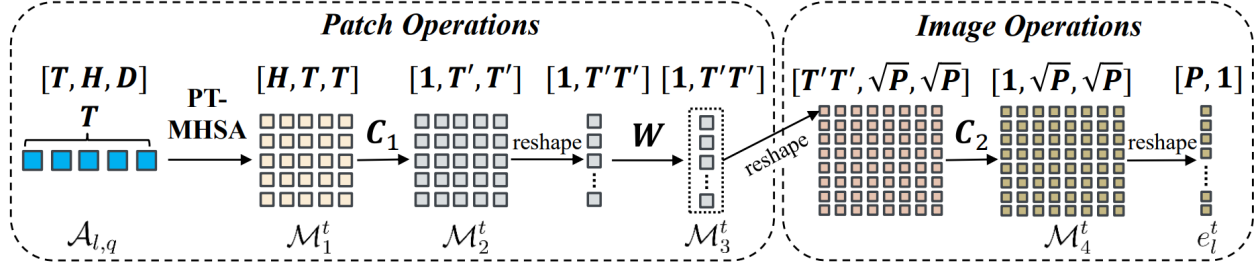


Figure 3. **Illustration of the Temporal Module Mechanism:** For enhanced clarity, we present a step-by-step demonstration of the operation within the temporal module. For illustration purpose, this example only considers a single attribute, where $\Gamma^t = \{\mathcal{A}_{l,q}\}$. For scenarios involving multiple attributes, simply replace H with $(|\Gamma^t| \times H)$.

dent Fully-Connected (FC) classification heads to predict the level of Deepfake in the temporal FC^t , spatial FC^s , and spatio-temporal FC^{st} branches. The overall Deepfake score is the average score of the respective heads.

3.2. Spatial Module

The spatial module in our decoder block is designed to capture manipulation cues from the extracted patch features $\mathcal{P}_l$. In the following, we first introduce the spatial module pipeline, then we describe the mining process used to extract layer-wise facial component attributes for the FCG. Finally, we detail the FCG loss used during training, which steers the spatial module’s focus towards key facial regions such as the lips, skin, eyes, and nose.

Pipeline. Each spatial module in the decoder block has its own set of N learnable parameters, denoted as $q_l^s \in \mathbb{R}^{N \times (H \times D)}$. These learnable parameters are fed into the Cross-Attention (CA) module as queries, while the keys are the target attribute γ^s of the video patches, and the values are the video patch embeddings. This process is expressed as $\mathcal{M}_1^s = \text{CA}(q_l^s, \mathcal{A}_{l,\gamma^s}, \mathcal{P}_l)$, where $\mathcal{M}_1^s \in \mathbb{R}^{T \times N \times (H \times D)}$, followed by averaging over the temporal and query dimensions of $\mathcal{M}_1^s$ and reshaping the tensor to obtain the layer-wise spatial feature $e_l^s \in \mathbb{R}^{(H \times D) \times 1}$.

Facial Attribute Mining. In this step, we collect layer-wise features associated with key facial parts (lips, skin, eyes, nose) from the CLIP image encoder to further guide the learnable queries q_l^s in the spatial module during training. First, we gather frames from facially aligned videos with random augmentations, such as resizing, cropping, and bidirectional flipping. These frames are then fed into the CLIP image encoder to extract the target layer-wise attributes $\mathcal{A}_{l,\gamma^s}$. Facial landmarks are detected for each frame, allowing us to collect the patch attributes corresponding to the facial part landmarks. The patch attributes for each facial part are averaged after L_2 -normalization, producing the layer-wise facial part attributes $\varphi_l^s \in \mathbb{R}^{4 \times (H \times D)}$, where the first dimension corresponds to the lips, skin, eyes, and nose, respectively. The

augmentations applied in the first step are crucial to remove the positional encoding in the features.

Facial Component Guidance. When applying the Facial Component Guidance (FCG), the number of parameters N in the spatial module is constrained to align with the number of facial parts ($1 \leq N \leq 4$). The FCG loss is defined as the InfoNCE loss between q_l^s and φ_l^s , encouraging each query in q_l^s to closely match its corresponding facial part attribute in φ_l^s . The FCG loss is computed as shown in Equation 1:

$$\mathcal{L}_{FCG} = \frac{1}{NL} \sum_{l=1}^L \sum_{i=1}^N \ln \left(\frac{\exp(\cos(q_{l,i}^s, \varphi_{l,i}^s) \times \tau)}{\sum_{j=1}^N \exp(\cos(q_{l,i}^s, \varphi_{l,j}^s) \times \tau)} \right), \quad (1)$$

where τ denotes the softmax temperature and $\cos(\cdot, \cdot)$ stands for the cosine similarity between two vectors. We apply the InfoNCE loss to contrastively guide the learnable queries, aiming to balance freedom between learning forgery cues and maintaining generalizability.

3.3. Temporal Module

Our temporal module in the decoder blocks focuses on capturing the temporal inconsistencies in Deepfake samples. As illustrated in Figure 3, it consists of four major steps, which are described below in detail.

For each fixed patch location, we apply Multi-Head Self-Attention (MHSA) along the temporal axis of the attributes in $\Gamma^t \subseteq \{\mathcal{A}_{l,q}, \mathcal{A}_{l,k}, \mathcal{A}_{l,v}\}$, we refer this process as Patch-Temporal Multi-Head Self-Attention (PT-MHSA). This step creates a temporal affinity map $\mathcal{M}_1^t \in \mathbb{R}^{P \times (|\Gamma^t| \times H) \times T \times T}$. We then apply the softmax function along the last axis to normalize the affinity values.

Next, we aggregate the multi-attribute, multi-head affinity scores at adjacent temporal locations using a 2D kernel $\mathbf{C}_1$. This kernel convolves over the last two axes of $\mathcal{M}_1^t$, with the second axis $(|\Gamma^t| \times H)$ as the input channel, producing a feature map $\mathcal{M}_2^t \in \mathbb{R}^{P \times T' \times T'}$.

To promote interaction across all temporal representations, we reshape the last two axes of $\mathcal{M}_2^t$ into a single axis with dimensions $T' \times T'$, then process the feature map

through a residual linear layer $\mathbf{W}$ to produce a new feature map $\mathcal{M}_3^t \in \mathbb{R}^{P \times (T' \times T')}$.

Finally, to aggregate with spatially adjacent patches, we reshape $\mathcal{M}_3^t$ into $\mathbb{R}^{(T' \times T') \times \sqrt{P} \times \sqrt{P}}$ and apply a 2D convolution kernel $\mathbf{C}_2$ to create the feature map $\mathcal{M}_4^t \in \mathbb{R}^{\sqrt{P} \times \sqrt{P}}$. This feature map is reshaped into $e_l^t \in \mathbb{R}^{P \times 1}$, serving as the temporal feature for the l -th decoder layer.

3.4. Objective Functions

The main objective comprises the focal loss [28] $\mathcal{L}_F(\cdot)$ of the three classification heads for temporal ($\mathcal{L}_T$), spatial ($\mathcal{L}_S$), and spatio-temporal ($\mathcal{L}_{ST}$) aspects:

$$\mathcal{L}_T = \mathcal{L}_F(\mathbf{FC}^t(\mathbf{LN}^t(E^t)), y), \quad (2)$$

$$\mathcal{L}_S = \mathcal{L}_F(\mathbf{FC}^s(\mathbf{LN}^s(E^s)), y), \quad (3)$$

$$\mathcal{L}_{ST} = \mathcal{L}_F(\mathbf{FC}^{st}(\mathbf{cat}(\mathbf{LN}^t(E^t), \mathbf{LN}^s(E^s))), y), \quad (4)$$

where $\mathbf{cat}(\cdot)$ represents the concatenation operation that merges the given features, E^s and E^t are spatial and temporal features aggregated from each layer, and y is the ground truth binary label. The overall objective during training is defined as:

$$\mathcal{L}_{total} = \mathcal{L}_T + \mathcal{L}_S + \mathcal{L}_{ST} + w\mathcal{L}_{FCG}, \quad (5)$$

with w adjusting the influence of the FCG loss.

4. Experiments

In the following sections, we present the implementation details, introduce the evaluation datasets and settings, and provide extensive experiments to demonstrate the effectiveness of our framework across various aspects, including generalizability, parameter efficiency, data efficiency, and robustness. For additional experiments please refer to the supplementary materials.

4.1. Implementation Details

We adopt a similar video pre-processing pipeline to RealForensics [13], which includes the following steps: 1) utilizing 2D-FAN¹ to extract facial landmarks from video frames, 2) aligning the detected faces with the mean face from the LRW [5] dataset, and 3) extracting the aligned faces from the videos. During training, non-overlapping clips ranging from 2 to 4 seconds are randomly extracted from each video, and 10 frames are uniformly sampled ($T = 10$) from the extracted clips. The sampled frames undergo a video-level augmentation pipeline, which includes horizontal flipping, random resizing and cropping, RGB shifting, hue and saturation adjustments, random modifications to brightness and contrast, image compression, blurring, and downscaling. This diverse augmentation strategy aims to simulate a wide range of real-world variations.

¹<https://github.com/1adrianb/face-alignment>

Model	Pub. / Mod.	CDF	DFDC	FSh	DFo	WDF
Xception [42]	ICCV'19 / Img.	73.7	70.9	72.0	84.5	-
Face X-ray [26]	CVPR'20 / Img.	79.5	65.5	92.8	86.8	-
LipForensics [12]	CVPR'21 / Vid.	82.4	73.5	97.1	97.6	-
FTCN [55]	ICCV'21 / Vid.	86.9	74.0	98.8	98.8	-
SBI(c23) [44]	CVPR'22 / Img.	92.9	72.0	-	-	-
RealForensics [13]	CVPR'22 / Vid.	86.9	75.9	99.7	99.3	-
AltFreezing [51]	CVPR'23 / Vid.	89.5	-	<u>99.4</u>	<u>99.3</u>	-
CADDMM [9]	CVPR'23 / Img.	<u>93.9</u>	73.9	-	-	-
TALL-Swin [52]	ICCV'23 / Vid.	90.8	<u>76.8</u>	99.7	99.6	-
Yan et al. [53]	CVPR'24 / Img.	91.1	77.0	-	-	-
Choi et al. [4]	CVPR'24 / Vid.	89.0	-	99.0	99.0	-
Hong et al. [14]	CVPR'24 / Img.	91.6	75.2	-	-	<u>73.4</u>
LAA-Net [33]	CVPR'24 / Img.	95.4	-	-	-	80.0
RealForensics*	CVPR'22 / Vid.	88.9	<u>75.8</u>	98.7	99.5	83.3
LAA-Net*	CVPR'24 / Img.	<u>93.6</u>	73.8	69.8	75.5	80.8
Ours	-	95.0	81.8	<u>98.1</u>	99.6	87.2

Table 1. **Generalizability Evaluation on Unseen Datasets:** The video-level AUC (%) of our method on unseen datasets (CDF, DFDC, FSh, DFo, and WDF) after training on the FF++ dataset. The scores from previous works are directly reported from their respective papers. Models marked with * are re-evaluated using the official code and model checkpoint under default settings. The publication and modality of each work are provided for reference.

Unless otherwise specified, our framework is based on the CLIP ViT-L/14² image encoder, which consists of 24 encoder layers ($L = 24$). In the spatial module, we set $\gamma^s = k$. For the temporal module, we set $\Gamma^t = \{\mathcal{A}_{l,q}, \mathcal{A}_{l,k}, \mathcal{A}_{l,v}\}$ and the convolution kernels $\mathcal{C}_1$ and $\mathcal{C}_2$ are configured with a kernel size of 5, padding of 2, and a stride of 1. We employ the AdamW optimizer, setting the learning rate (lr) to 1e-4, with β parameters set to (0.9, 0.999) and a weight decay of 1e-3. The weight (w) of the loss function is set to 0.15. We utilize focal loss with a gamma value of 4 to focus on the challenging samples. To optimize training, we implement an early stopping strategy, monitoring the Area Under the Receiver Operating Characteristic (AUROC) score on the validation dataset with a patience of 10 epochs. Our training process spans 30 epochs using $4 \times \text{V100}$ GPUs, with a batch size of 60 clips per GPU. **The overall training process is completed within approximately 16 GPU hours.**

4.2. Evaluation Settings for Deepfake Detection

Aligning with all prior work in Deepfake detection is challenging due to the diversity of evaluation protocols. For fair comparison, we primarily follow the evaluation protocols established by RealForensics [13]. The datasets used in our experiments are as follows: The **FaceForensics++**

²<https://github.com/openai/CLIP>

Comp.	CDF	DFDC	FSh	DFo	WDF	Avg.
Ours	95.0	81.8	98.1	99.6	87.2	92.3
w/o S mod.	86.2	75.7	98.3	99.4	75.2	87.0(-5.3)
w/o T mod.	91.0	83.0	87.0	98.0	<u>86.1</u>	89.0(-3.3)
w/o FCG	<u>92.0</u>	81.7	96.3	99.1	85.2	90.9(-1.4)
w/o focal	91.9	81.5	<u>98.1</u>	99.0	85.5	<u>91.2</u> (-1.1)

Table 2. **Ablation Study:** The components are the spatial module (S mod.) and the temporal module (T mod.) within the decoder block, the proposed FCG loss (FCG) and the focal loss (focal).

w/o	Model	FF++	CDF	DFDC	FSh	DFo	WDF	Avg.
DF	R.F.	100.0	69.2	72.4	90.5	99.2	54.2	77.1
	Ours	99.6	93.0	79.1	97.1	99.2	83.1	90.3
FS	R.F.	97.1	80.0	71.2	98.5	99.4	78.8	85.6
	Ours	93.2	93.2	75.9	96.3	99.5	86.8	90.3
F2F	R.F.	99.7	75.7	72.3	99.1	99.9	83.1	86.0
	Ours	99.8	92.5	81.1	96.6	99.0	82.3	90.3
NT	R.F.	99.2	80.8	72.4	93.8	98.6	68.3	82.8
	Ours	94.6	92.4	82.3	96.4	99.2	84.4	90.9

Table 3. **Extended FF++ Leave-One-Out (LOO) Evaluation:** In this experiment, we compare with RealForensics (R.F.) by extending the FF++ LOO evaluation to include cross-dataset assessment. The R.F. performance on the FF++ dataset is directly reported from the paper, while the scores for cross-datasets are evaluated using the LOO weights provided by the official repository.

(FF++) dataset [42] contains 1,000 real videos and 4,000 Deepfake videos generated with face-swapping techniques (Deepfake [7], FaceSwap [24]) and reenactment methods (NeuralTexture [47], Face2Face [46]). The **CelebDF-v2 (CDF)** [27] dataset includes 5,639 high-quality DeepFake videos of celebrities. The **DeepFake Detection Challenge (DFDC)** [8] dataset is a large-scale dataset containing 124,000 Deepfake videos generated with eight different algorithms. The **FaceShifter (FSh)** [25] dataset produces high-fidelity Deepfake videos using an occlusion-aware face swapping algorithm. The **DeeperForensics (DFo)** [17] dataset comprises 60,000 videos with generic perturbations to enhance data diversity. Lastly, the **WildDeepfake (WDF)** [57] dataset includes 7,314 face sequences from 707 Deepfake videos sourced from the internet. We use the complete FF++ dataset for our experiments, but we only use the test split of the other datasets for evaluation purpose.³

Unless otherwise stated, the experiments follow the given regime: 1) The models are trained on the training subset of the High Quality (HQ) compression FF++ dataset. 2) The reported scores are presented in video-level AUROC

³Following RealForensics, we use only a subset (3,814 samples) of the DFDC test split (5,000 samples) for evaluation.

Method	# T.P.	CDF	DFDC	FSh	DFo	WDF	Avg.
LCP	2.0K	75.8	77.0	88.0	92.4	82.1	83.3
VPT _{Shallow}	3.1K	78.9	76.4	89.0	92.7	82.9	84.0
VPT _{Deep}	26.6K	81.1	<u>77.5</u>	<u>90.0</u>	<u>98.3</u>	<u>86.7</u>	<u>86.7</u>
EVL	58.3M	80.3	75.8	87.7	96.5	79.1	83.9
FFT	303.0M	<u>88.3</u>	75.2	84.8	93.8	83.8	85.1
Ours	250.0K	95.0	81.8	98.1	99.6	87.2	92.3

Table 4. **Comparison of Model Architectures:** We compare our adaptation methods with the various adaptation methods under the cross-dataset evaluation. The models are based on the ViT-L/14 CLIP image encoder. We also present the number of Trainable Parameters (# T.P.) for comparison. 4.1.

#Data	CDF	DFDC	FSh	DFo	WDF	Avg.
100%	95.0	81.8	98.1	99.6	87.2	92.3
75%	<u>93.1</u>	<u>81.4</u>	<u>97.7</u>	99.3	<u>85.5</u>	<u>91.4</u>
50%	92.0	<u>81.4</u>	96.2	<u>99.4</u>	84.7	90.7
25%	90.3	77.2	95.6	99.1	80.1	88.5
10%	78.3	73.3	91.9	99.0	75.2	83.5

Table 5. **Generalizability on Proportional Training Dataset:** To validate the efficiency of our approach on training samples, we assess the generalizability of our model trained on varying proportions of the training dataset. This evaluation demonstrates the model’s performance in scenarios with limited training samples.

in percentage. 3) Values in **bold** represent the best results, while those underlined indicate the second-best results.

4.3. Generalization to Unseen Dataset

In Table 1, we assess the generalization capability of our model trained on the FF++ dataset against five unseen datasets: CDF, DFDC, FSh, DFo and WDF. We include scores from previous methods as reported in their respective papers and official repositories. By integrating CLIP’s robust generalizability with our proposed framework, we observe notable improvements while facing unseen samples, outperforming the SOTAs on most datasets. This improvement is particularly pronounced in the challenging CDF, DFDC and WDF datasets, where our method establishes a significant lead of 1.4%, 6% and 3.9%, respectively.

4.4. Ablation Study on Model Components

In Table 2, we evaluate the contribution of various components to the framework’s generalization capability across different datasets. Our experiment highlights the crucial role of our dual-module architecture: removing either the spatial or temporal module results in an average performance decline of 5.3% and 3.3%, respectively. Notably, excluding the spatial module significantly impacts performance on the challenging CDF, DFDC, and WDF datasets,

Method	Dataset	Clean	CS	CC	Block	Noise	Blur	JPEG	Comp.	Avg
RealForensics	FF++ [42]	99.8	99.8	99.6	98.9	79.7	95.3	98.4	97.6	95.6 (-4.2)
Ours [†]		99.2	97.9	97.5	98.1	77.1	94.3	94.1	93.4	93.2 (-6.0)
RealForensics*	CDF [27]	79.3	71.5	64.8	68.3	58.8	56.6	62.9	63.4	63.8 (-15.5)
Ours [†]		91.0	89.4	87.6	90.1	74.6	77.9	80.1	75.7	82.2 (-8.8)

Table 6. **Video Robustness Evaluation:** The upper part of the table reports the robustness evaluation on FF++, while the lower part reports the evaluation on CDF. We follow the training protocols mentioned in Section 4.7 and denote the model as Ours[†]. We report the RealForensics’ FF++ robustness evaluation directly from the paper, while the robustness evaluation on CDF is conducted with the official checkpoint (denoted as RealForensics*).

underscoring the importance of facial semantic learning in accurately assessing video authenticity for these datasets. Additionally, removing the temporal module leads to a marked performance decrease on the FSh dataset, indicating the presence of temporal inconsistencies that provide important cues in this dataset. Our strategy of deploying independent modules to address both spatial and temporal aspects has proven effective for enhancing generalization.

Furthermore, we observe that using focal loss can slightly improve generalizability compared to cross-entropy loss, likely due to the focal loss’s ability to handle hard samples. Our proposed Facial Component Guidance (FCG) further enhances generalization by directing the layer-wise queries to focus on primary facial regions, yielding an average performance boost of 1.4%. The integration of these components collectively results in a significant improvement in the model’s overall performance.

4.5. Data-Efficiency Evaluation

We further assess the generalizability of our framework under additional constraints on the training dataset, such as excluding certain types of manipulations and limiting the size of the dataset. These constraints evaluate the data efficiency of our framework in transferring knowledge in the CLIP image encoder to the Deepfake detection task.

Constraint on FF++ Manipulation Types. In Table 3, we follow the Leave-One-Out (LOO) protocol to evaluate the framework’s performance on each of the four manipulation types in FF++ after training on the remaining three types. The results are presented under the FF++ column. Additionally, we assess the generalizability of the LOO model to unseen datasets (CDF, DFDC, FSh, DfO and WDF).

When evaluated on the FF++ dataset, RealForensics generally outperforms our proposed framework. However, RealForensics fails to maintain its performance in cross-dataset evaluations. In contrast, our approach consistently excels on the challenging CDF, DFDC, and WDF datasets and performs comparably on the FSh and DfO datasets. Since the DfO and FSh datasets use real videos from FF++ to create their Deepfake samples, we speculate that RealForensics may be biased towards FF++-specific cues for Deepfake detection. This could explain its stable perfor-

mance on the FSh and DfO datasets, despite the significant performance drop when tested on the completely unseen CDF, DFDC, and WDF datasets.

Constraint on Size of the Training Dataset. In this experiment, we scale the size of the FF++ training dataset to evaluate the framework’s efficiency with a limited number of Deepfake samples. The results are presented in Table 5, where the framework is trained on a portion of the dataset and then evaluated on the unseen datasets. Notably, our method exhibits robust performance, showing no significant decrease when trained on 75% of the dataset. Furthermore, it surpasses state-of-the-art methods on most datasets even when trained with only 50% of the dataset.

4.6. Evaluation on Parameter Efficiency

In Table 4, we highlight the effectiveness and parameter efficiency of our proposed framework on adapting the CLIP image encoder for Deepfake detection, we compare our framework with various adaptation methods: the Full Fine-Tuning (FFT) method, the Linear Classifier Probing (LCP) [1] method, the Visual Prompt Tuning (VPT) [16] method and the Efficient Video Learner (EVL) [30] framework. We apply both the ‘Deep’ and ‘Shallow’ versions of VPT, where VPT_{Deep} introduces one trainable prompt per layer in the image encoder, while VPT_{Shallow} introduces a single trainable prompt only at the first layer. Our framework is similar to EVL in deploying a side-network video learner that progressively decodes per-layer features. However, our framework employs two distinctive components with strong inductive bias – the spatial and temporal modules – unlike EVL, which uses an unified module to optimize all in a box. The experiment reveals our approach significantly outperforms previous adaptation methods across all datasets. Furthermore, our framework requires significantly fewer trainable parameters compared to EVL, being ≈ 200 times smaller. This highlights our method’s parameter efficiency and effectiveness in adapting the CLIP image encoder for Deepfake detection.

4.7. Robustness Evaluation Against Perturbations

To assess the robustness of our framework against common perturbations, we follow the protocol described in Real-

Forensics [13], training on the FF++ dataset with random flipping and random resize cropping augmentations. We then evaluate the model’s performance on the FF++ testing subset with perturbations introduced by DeeperForensics (DFo) [17]. These perturbations vary across five intensity levels and affect the image in terms of color saturation (CS), color contrast (CC), block-wise occlusions (Block), color component noise (Noise), Gaussian blur (Blur), JPEG compression (JPEG), and video compression (Comp.). The results are summarized in the upper section of Table 6.

Our framework appears less robust than RealForensics on the FF++ dataset, showing an average performance drop of 1.8% more when handling samples with perturbations. However, we note that RealForensics demonstrates strong affinity for the FF++ dataset, as discussed in Section 4.5. To provide a more comprehensive evaluation, we extend the robustness assessment to the unseen CelebDF (CDF) [27] dataset. We evaluate our model and the checkpoint provided by RealForensics⁴ on the CDF dataset with DFo-introduced perturbations applied. The results are presented in the lower section of Table 6. In this experiment, RealForensics struggles to maintain robustness when transitioning from FF++ to CDF, with an average performance drop of 15.5%. In contrast, our framework exhibits a relatively stable performance with an average drop of only 8.8%, demonstrating greater robustness across different datasets.

4.8. Attention Visualization of the FCG

In Figure 4, we present a comparative visualization of the per-frame affinity maps for three models: the fully fine-tuned CLIP image encoder (as shown in Table 4) and our framework with and without the FCG. The results presented are averaged over 100 randomly sampled clips from the FF++ testing dataset (please refer to Fig. 5 for visualizations of individuals). The upper part of the figure shows the averaged result when multiple queries are present in the model, while the lower part specifically displays each of the four queries in our framework with the FCG applied. This juxtaposition highlights the effectiveness of our strategy in directing the model’s attention toward four key facial regions, whereas the other models tend to focus on arbitrary small areas that lack generalizability for detecting Deepfakes.

5. Conclusion

In this paper, we present a novel PEFT framework that leverages the CLIP image encoder to detect Deepfake videos by identifying temporal inconsistencies and spatial artifacts in critical facial regions. We introduce the Facial Component Guidance (FCG) loss, which directs the spatial branch to focus on four key facial areas: skin, nose, eyes, and lips, for enhanced feature extraction. In cross-dataset

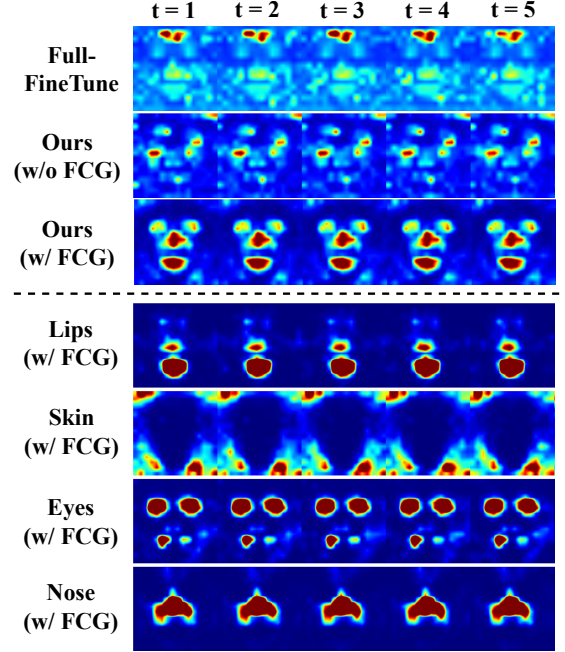


Figure 4. **Attention Visualization:** The upper section displays the per-frame affinity maps for three Deepfake detectors: the fully fine-tuned CLIP image encoder and our proposed framework with and without the FCG. The lower section further illustrates the per-frame affinity maps for each of the four queries in our framework with the FCG applied. The results demonstrate the effectiveness of our proposed FCG mechanism in steering the model’s attention toward different facial components.

evaluations, our method outperforms previous state-of-the-art methods across multiple datasets. In the experiments, we also reveal that prior evaluation methods may overlook the model’s bias toward the training dataset. Through extensive experiments, our method demonstrates advantages in parameter efficiency, data efficiency, and robustness.

6. Limitations and Future Work

In our framework, we sparsely sample a limited number of frames from each video clip to reduce memory consumption. However, this approach may overlook high-frequency artifacts present in Deepfake videos, potentially leading to incorrect predictions. Despite achieving commendable robustness compared to other methods, we believe that performance could be further improved by enhancing robustness during the pretraining stage. Our framework is designed with high extensibility, allowing for the deployment of different modules in the decoder block to address various aspects of Deepfake detection. Incorporating self-supervised learning methods as additional modules in the decoder block could also be a promising avenue for further development and improved performance.

⁴Model checkpoint from RealForensics.

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